

Expanded Form, Comparing, and Rounding

Answer Key

Grade 4–5 Multi-Digit Place Value and Rounding

Quick Answers

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|----------|----------------------------|
| 1. 6382 | 7. — |
| 2. 7000 | 8. 100 |
| 3. 6400 | 9. 26000 |
| 4. — | 10. 70425, 72045, 72405 |
| 5. 43065 | 11. 9000 |
| 6. 47000 | 12. —, 38472, 38000, 38000 |
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Curriculum

Level: Grade 4–5 Multi-Digit Place Value and Rounding

4.NBT.A.1–A.3 / 5.NBT.A.1–A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 16 tagged checks.

1. Write 6382 in expanded form.

Step 1. Expanded form breaks a number into what each digit is worth, so name the place of each digit first: 6 thousands, 3 hundreds, 8 tens, 2 ones.

Step 2. Turn each into its value: $6 \times 1000 = 6000$, $3 \times 100 = 300$, $8 \times 10 = 80$, $2 \times 1 = 2$.

Step 3. Write the values as a sum:

$$6382 = 6000 + 300 + 80 + 2$$

Check: adding the four values returns the original number. That is the test for any expanded form.

2. Value of the digit 7 in 47308.

Step 1. “Value” asks what the digit is worth where it sits, not what digit it is. Count places from the right: ones, tens, hundreds, thousands — the 7 is in the thousands place.

Step 2. 7×1000 .

$$7000$$

Common wrong answer: 7 — that is the digit, not its value.

3. Round 6382 to the nearest hundred.

Step 1. Rounding to the hundreds means choosing between the hundred below (6300) and the hundred above (6400), so look at the digit one place to the right — the tens digit.

Step 2. The tens digit is 8, which is five or more, so round up.

Step 3. The hundreds digit goes from 3 to 4 and everything to its right becomes zero.

$$6400$$

Circle: the 8 in the tens place is the digit that decided it.

4. Compare 47308 and 47380.

Step 1. Start at the largest place and move right only while the digits match: ten-thousands $4 = 4$, thousands $7 = 7$, hundreds $3 = 3$.

Step 2. The tens place is the first difference: 0 against 8. That decides the whole comparison, whatever comes after it.

Step 3.

$$47308 < 47380$$

The deciding place: the tens.

5. Standard form of $40000 + 3000 + 60 + 5$.

Step 1. Each addend names one place. Sort them by size: 4 ten-thousands, 3 thousands, 6 tens, 5 ones — and notice there is no hundreds addend at all.

Step 2. A missing place does not disappear; it needs a zero to hold the other digits in position.

Step 3. Writing the digits in order — ten-thousands, thousands, hundreds, tens, ones — gives

$$43065$$

Common wrong answer: 4365 — the placeholder zero was left out, and every digit to its left slid one place down in value.

6. Round 47308 to the nearest thousand.

Step 1. Rounding to thousands is a choice between 47000 and 48000, so read the digit one place to the right: the hundreds digit.

Step 2. The hundreds digit is 3, which is four or less, so the thousands digit stays where it is.

Step 3. Everything to the right of the thousands place becomes zero.

$$47000$$

The digit that decided it: the 3 in the hundreds place.

7. Kai wrote $8090 > 8900$. Correct him.

Step 1. Comparing is decided by the largest place where the digits differ — not by which number contains the biggest digit anywhere.

Step 2. Thousands: $8 = 8$, so move right. Hundreds: 0 against 9. That is the first difference, and 0 is less than 9.

Step 3.

$$8090 < 8900$$

Kai should have compared the *hundreds* place first, once the thousands matched. A 9 in the tens place is worth far less than a 9 in the hundreds place — position beats size of digit.

8. In 55505: how many times greater is the ten-thousands 5 than the hundreds 5?

Step 1. Both digits are 5, so the question is entirely about their places. Compare the two values directly by dividing.

Step 2. $50000 \div 500$. Both end in two zeros, so this is $500 \div 5 = 100$.

$$100$$

Why it comes out as a power of ten: the two places are two steps apart, and each step to the left multiplies by ten — so $10 \times 10 = 100$.

9. Round 25649 to the nearest thousand.

Step 1. The choice is between 25000 and 26000, so read the hundreds digit.

Step 2. The hundreds digit is 6, which is five or more, so the thousands digit rounds up from 5 to 6.

$$26000$$

Common wrong answer: 25000 — from reading the 4 in the tens place instead of the 6 in the hundreds. Only the digit immediately to the right of the rounding place matters.

10. Order 72405, 72045, 70425 from least to greatest.

Step 1. All three use the same digits, so nothing but place value separates them. Compare from the left.

Step 2. Thousands place: 72405 and 72045 both have 2 thousands; 70425 has 0. So 70425 is the smallest.

Step 3. Between the other two, the hundreds place decides: 4 against 0, so 72045 comes before 72405.

Step 4. Least to greatest:

70425, 72045, 72405

Why this is a challenge problem: a student comparing digit by digit without tracking places will read all three as “the same numbers”.

11. Round 8950 to the nearest hundred.

Step 1. The choice is between 8900 and 9000, so read the tens digit.

Step 2. The tens digit is 5, so round up: the hundreds digit 9 must become 10 hundreds.

Step 3. Ten hundreds is one thousand, so the carry moves left: the 8 thousands become 9 thousands and the hundreds place becomes 0.

9000

The thousands digit changed because rounding the hundreds up past 9 has nowhere to go inside the hundreds place — ten hundreds *is* a thousand, so the extra one carries into the next place, exactly like a carry in addition.

12. Challenge: Friday 38472, Saturday 38247.

Step 1 (a). Name each digit’s place in Friday’s number — 3 ten-thousands, 8 thousands, 4 hundreds, 7 tens, 2 ones — then write the values as a sum.

$38472 = 30000 + 8000 + 400 + 70 + 2$

Step 2 (b). Compare from the left: ten-thousands match, thousands match, and the hundreds place is the first difference, 4 against 2.

$38472 > 38247$

Friday had more people.

Step 3 (c). Round each to the nearest thousand by reading the hundreds digit. Friday: 4, so the thousands stay. Saturday: 2, so the thousands stay. Both land on the same value.

38000

Step 4 (d). *Sample answer (accept any equivalent explanation):* “Both days round to 38000 because rounding throws away everything below the thousands place, and the two days only differ in the hundreds and tens. To settle which day had more people I would compare the exact numbers, or write them in expanded form and compare the hundreds.”

Grading note for part (d): look for the idea that rounding removes the very places where the two numbers differ. This is the one item on the sheet that is not machine-checked.